

Software Requirements Specification

for

Blood Donation Management System

Prepared by

Seren Moonesawmy	2516246
Ridwaan Bin Farook Soopun	2516894
Lokeshav Anand Makhun	2514204
Baboo Jatinsingh Gowreesunkur	2516213
Ayush Liladhar Ramrooch	2516786

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1. Introduction

1.1 Aims & Objectives

Our Blood Donation Management System (BDMS) allows clients — blood banks and patients — to search for donor records instantaneously whenever information is required. Our BDMS maintains blood inventory and alerts donors in any case of emergencies based on their geo-location using a KNN algorithm. Our aim is to ensure efficient management of blood donors and to monitor the donation process itself to save lives. Every 2 seconds someone needs blood. We also ensure the privacy of our users by encrypting all data and giving specific access rights to each user of the system.

1.2 SRS team members and Sub-system worked on

Student name	Student ID	Subsystem worked
Seren Moonesawmy	2516246	Recipient Management
Ridwaan Bin Farook Soopun	2516894	Blood Inventory Management
Lokeshav Anand Makhun	2514204	Donor Management
Baboo Jatinsingh Gowreesunkur	2516213	Reporting and Analytics
Ayush Liladhar Ramrooch	2516786	User Access Management

1.3 Recommended Software Process Model

Our recommended software process model for the Blood Donation Management System is the **Incremental and Iterative Model** based on the **Scrum framework** (Agile methodology). Since, the system is created and then given into small but usable increments -- the product is delivered before the end. Thus, we use an incremental model. Each increment is reviewed first and then tested. It is improved where possible according to feedback received before developing the next increment.

Since the Blood Donation Management system has changing requirements, the agile methodology can easily adapt by updating backlogs and managing sprints. Moreover,

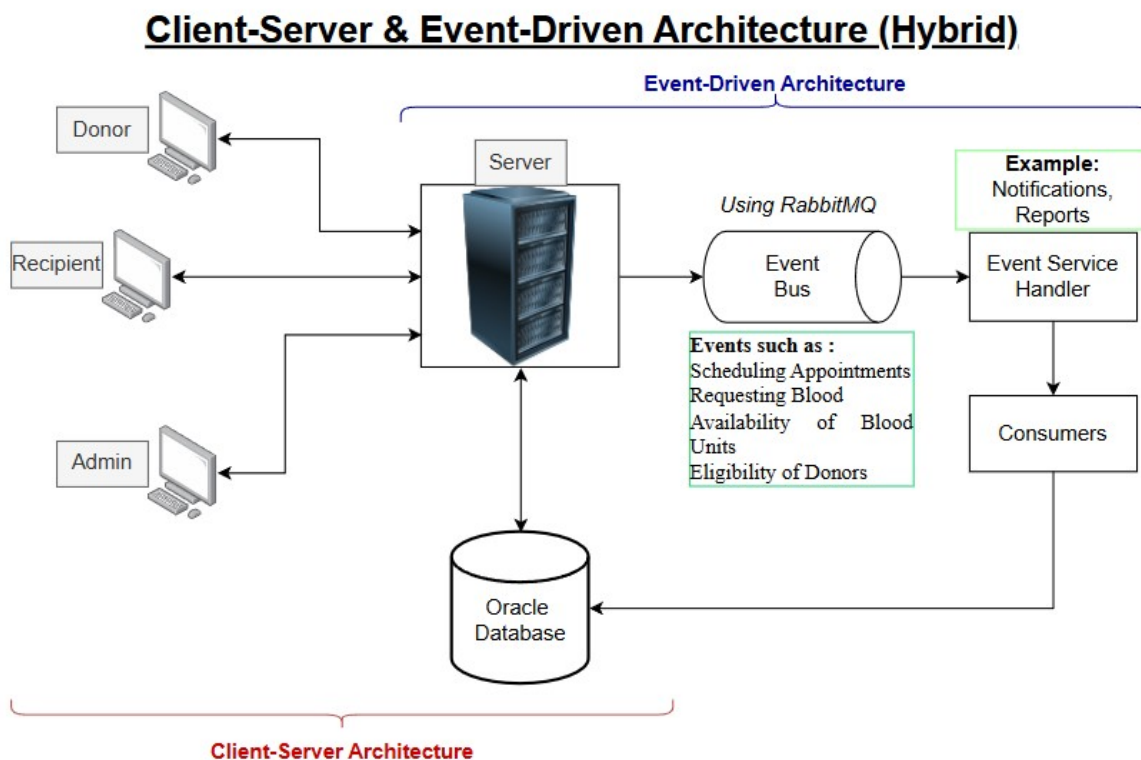
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hospitals or blood banks can update their requirements for new or improved blood testing protocols. This model allows health authorities, doctors, staff, and donors to provide their feedback after a sprint.

2. Overall Description

2.1 System Architecture Diagram

Our Blood Donation Management System is designed using a hybrid system architecture, by seamlessly combining elements of both of **Client-Server Architecture** and **Event-Driven Architecture** to achieve efficiency, security, scalability and responsiveness.

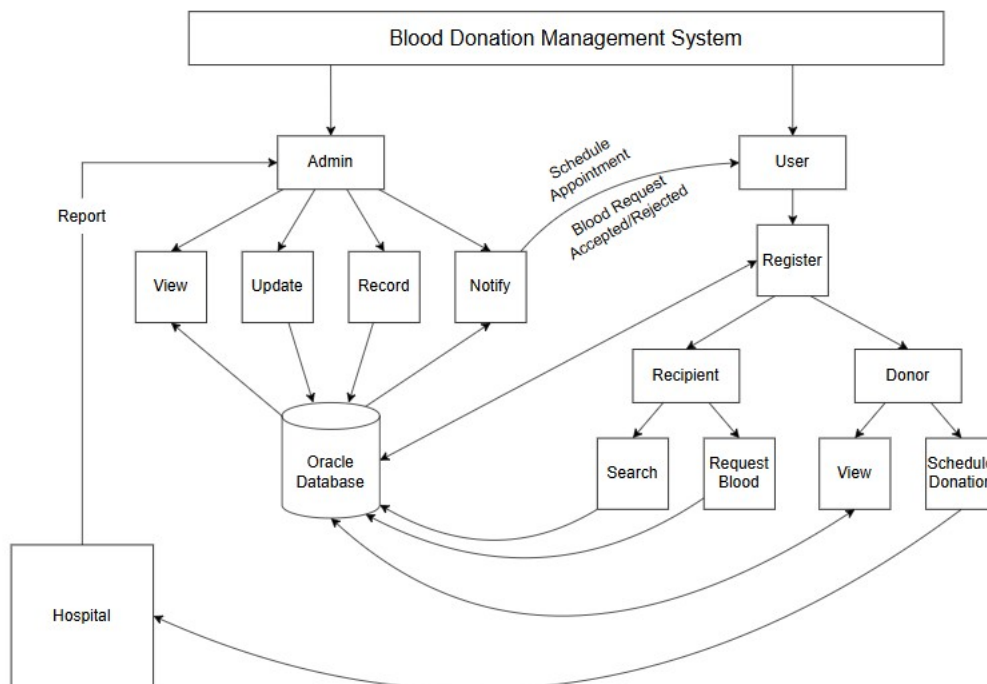


In the **Client-Server Architecture**, clients (web application programs) interact with a central server that manages requests, processes data and communicates with the database. This structure ensures centralized control, secure data management, easy maintenance and scalability which is crucial for designing this critical system. The server handles core operations which include donor management, recipient management, blood inventory management, report generation and other administration processes while the clients provide an intuitive interface for users mentioned above (donor, hospitals, administrators).

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In the **Event-Driven Architecture**, the system can automatically respond to specific events or triggers. For instance, after users have requested for blood units, the system can automatically check for its availability and send notifications to relevant users. Moreover, if a donor has scheduled an appointment, the system can instantly check his eligibility and donation history after which a suitable notification/report is sent. This process relies on the producers (donor, recipient, admin from the server generating an event), the event bus (routing events from producers to consumers), event service handler (how to respond to possible events), and consumers (ensures who listens for and responds to those events). This model enhances system responsiveness, reduces manual intervention and provides real-time updates.

By integrating both architectures, the BDMS benefits from the robust data management and security of the client-server model, while still leveraging the real-time and extremely responsive capabilities of event-driven design. This hybrid approach contributes to the system being reliable scalable and user-friendly to meet the operational needs of the BDMS, while managing and securing this sensitive and critical system.



2.2 Subsystem Description

2.2.1 Donor Management

The Donor Management system shall allow new donors to register for an account by entering their personal details (name, address, date of birth, sex, NID, loginID, password) and their authentic medical records, which are sent to the database. It shall also allow registered users to log in and view their profile along with their donation history including dates and blood units. Users can also request to change their personal information for any discrepancy but not their medical records. The system shall allow them to schedule their appointment based on their donation history and medical report from the hospital as well as to choose the type of donation (whole blood, power red, platelets and plasma).

2.2.2 Recipient Management

The recipient management subsystem is responsible for handling different requests and processes related to patients and hospitals concerning blood units. It links hospitals (recipients) and blood bank inventory. The subsystem ensures that mainly authorized personnel and hospitals can submit blood requests, which are checked out and matched with available stocks and tracked until completion. Any emergency is supported through transparency, accuracy, and speed.

2.2.3 Blood Inventory Management & Search Module

The Blood Inventory Management & Search Module system will automate the tracking of blood products from collection to administration. It shall also monitor blood stock levels, availability, and expiry dates.

2.2.4 Reporting and Analytics

The Reporting and Analytics subsystem is tasked to deliver periodic analytical reports on donors, recipients, blood donations campaigns, and blood bank inventory. It will ensure systematic management of the blood stock and provide valuable insight into donation patterns and inventory levels. The user will be able to determine the success rate of each donation campaign by comparing it with set targets. This subsystem will serve as a pillar for the evolution of the system. The sorted data will enable users to make informed decisions and to implement future operations such as blood collections... more effectively and efficiently.

2.2.5 User and Access Management, Database and Administration Subsystem

The User and Access Management system will allow a donors, admin staffs and patients to create a login profile. The system has a centralized database of store all details about blood banks, hospitals, admin officers, patients, and donors. This will enable doctors and recipients to know if the required blood is available. Furthermore, it will enable donors to know if their blood can be accepted or not. This subsystem is interlinked with all the other subsystems.

Different users will have different access rights. For instance, patients will be able to only view of his/her own personal details and only view medical reports. Administrators such as doctors and nurses will have access to the medical records of each of their patients and details about them.

3. User and System Requirements

User requirements

- U1** The user (donor) shall be able to create a profile account.
- U2** The user (donor) shall be able to update their user profile.
- U3** The user (recipient) shall be able to only view his/her medical records.
- U4** The user (donor) shall be able to donate blood.
- U5** The user (recipient) shall be able to input their credentials.
- U6** The user (recipient) shall be able to request blood.
- U7** The user (recipient) shall be informed of request status.
- U8** The user (recipient) shall be able to check instantly if compatible (transfusable) blood is available or not.
- U9** The user (recipient) shall be able to check the location of nearest available donor if none of the inventories have matching blood.
- U10** The user (recipient) shall be routed to the nearest inventory which has matching blood.
- U11** The user (Admin) shall be provided with a report for all donors.
- U12** The user (Admin) shall be provided with a report of all recipients.
- U13** The user (Admin) shall be provided with reports for blood units.
- U14** The user (Admin) shall be provided with reports for each blood donation campaign.
- U15** The user (donor, recipient & Admin) shall be allowed to create their own login credentials.
- U16** The user (donor, recipient & Admin) shall be protected from data loss.

System Requirements

3.1 Functional Requirements

A. Donor Management

- A1** The system shall prompt the user to enter his name to create an account.
- A2** The system shall prompt the user to enter his address.
- A3** The system shall prompt the user to enter his gender.
- A4** The system shall prompt the user to enter his date of birth.
- A5** The system shall prompt the user to enter his National ID (NID).
- A6** The system shall prompt the user for a loginID for his account.
- A7** The system shall prompt the user for a password for his account.
- A8** The system shall send all the user's input details to the database.
- A9** The system shall prompt the registered user for their loginID and password to login to their account.
- A10** The system shall send the credentials to the database to request access.
- A11** The system shall display error messages if access is not granted.
- A12** The system shall display error messages for empty records (such as not entering blood group, address, eligibility)
- A13** The system shall display the user's profile* after logging in.
- A14** The system shall allow the user to request a change in their profile.
- A15** The system shall allow the user to modify their user details if request is granted.
- A16** The system shall not allow the deletion of medical records of a user.
- A17** The system shall not allow the modification of medical records by the user.
- A18** The system shall allow a donor to schedule an appointment to donate blood at a registered hospital.

*A profile is created by the database based on the donor details. It includes name, age, sex, NID and date when the account was created.

B. Recipient Management

- B1** The system shall allow hospitals and clinics to register their credentials.
- B2** The system shall allow only authorized users to access the recipient portal.
- B3** The system shall let users create new blood requests, taking into consideration blood type, quantity and urgency.
- B4** The system shall validate requests before submission.
- B5** The system shall allow the viewing of requests in real time.
- B6** The system shall send a notification to users based on their request state.
- B7** The system shall generate a monthly report of the number of allocations and requests made in hospitals.
- B8** The system shall check the inventory for the availability of blood.
- B9** The system shall match blood units automatically.

C. Blood Inventory Management & Search Module

- C1** The system shall provide geolocation-based identification of donors.
- C2** The system shall make use of First-In-First-Out (FIFO) stock management.
- C3** The system shall provide a real-time blood compatibility checker, which automatically validates blood types against request parameters.
- C4** The system shall automatically refresh the stock in real time in case of a successful donation.
- C5** The system shall provide Cross-Inventory Blood Request Routing.
- C6** The system shall automatically search nearby inventories and propagates the request if the local inventory of a hospital does not have the requested blood type.

D. Reporting and Analytics

- D1** The system shall generate an annual report for all donors recorded.
- D2** The system shall generate an annual report for all active donors.
- D3** The system shall provide a report of all recipients recorded.
- D4** The system shall provide a report for all active recipients.
- D5** The system shall provide a report of the total blood units collected monthly.
- D6** The system shall provide a report of the total blood units utilised monthly.
- D7** The system shall provide a report of the total blood units discarded monthly.
- D8** The system shall provide a report for the reasons of discarding blood units.
- D9** The system shall generate a list for blood units which are expiring within 30 days.
- D10** The system shall generate a report for each blood type in the inventory.
- D11** The system shall generate a report for each blood donation campaign detailing the number of donors.
- D12** The system shall generate a report for each blood donation campaign detailing the total blood collected by type.
- D13** The system shall generate a report detailing the blood types sought by recipients, sorted by increasing demands.
- D14** The system shall allow a registered user to login with their user account.
- D15** The system shall generate a report for each disease type detected in blood tests over a specified period.

E. User and Access Management, Database and Administration Subsystem

E1 The system shall allow a patient to only access the viewing of his/her own personal details.

E2 The system shall allow a patient to only access the viewing of his/her own medical details.

E3 The system shall allow administrators (doctors, nurses & IT staffs) to access the medical records of each of their patients.

E4 The system shall allow administrators (doctors, nurses & IT staffs) to access the personal details of their patients.

E5 The system shall allow administrators (IT staffs) to grant specific privileges to different users of the system.

E6 The system shall allow donors, recipients and admins to enter their login credentials.

E7 The system shall store information about recipients, donors, donations, blood inventory, and requests in a centralized database.

E8 The system shall allow automatic weekly backups.

E9 The system shall allow the storage of reports generated.

3.2 Non-Functional Requirements

- The system shall have a centralized database of all blood banks, hospitals, admin officers, patients and donors.
- The system shall be available 24/7
- The system shall work on Oracle, PostgreSQL and MySQL database.
- The system shall support at least 1000 concurrent users without performance degradation.
- The system shall respond to any user request within 3 seconds under normal load.
- The system shall complete database queries within 2 seconds.
- The system shall be able to handle up to 10,000 donor records efficiently.
- The system shall have an uptime of 99.5% during operational hours.
- The system shall recover within 5 minutes in case of a system failure.
- The system shall perform data backups automatically every 24 hours.
- The system shall encrypt all user data
- Users must log in with secure credentials (username and password with a two-factor authentication).
- The system must comply with data protection and privacy laws.
- The user interface should be intuitive and easy to navigate for all users.
- The system shall support English and French languages.
- All forms should include clear instructions and validation messages for incorrect inputs.
- The database should be scalable to accommodate an increasing number of users and donation records.
- The system should have a modular architecture to simplify troubleshooting and upgrades.
- The application should be compatible with Android, iOS platforms Windows and Linux
- Every transaction (donation, request, blood test update) must be logged with date, time, and user ID.
- Logs should be retained for at least one year for audit purposes
- The system shall trace all different requests and make sure they are compliant.

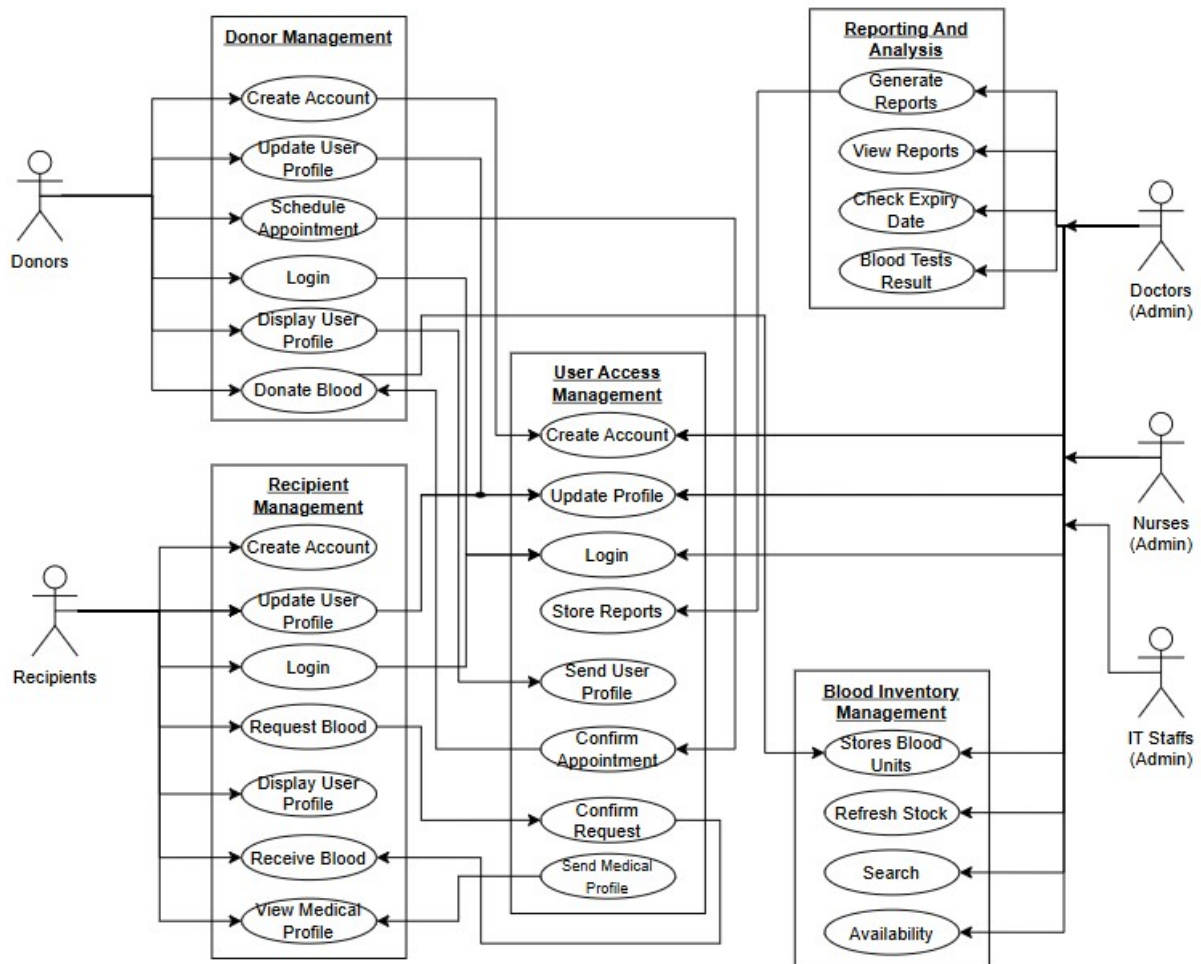
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- The system shall follow health data privacy regulations.
- The system shall be written in Java.
- The system shall allow recording up to 10000 entries a day.
- The system shall take 5 seconds to store and process each entry.
- The system shall retrieve and display up to 100 registered donor details within 2 seconds after executing the search query.
- The system shall store details locally if it cannot connect to the database.
- The system shall allow authorized users to access the interface within 5 seconds after logging in.
- The system shall ensure that stock levels are automatically synchronized across all connected hospitals and blood banks.
- The system shall Integrate machine learning algorithms to improve forecasting of blood demand.
- The system shall make use of a cloud-based, smart blood inventory management system.
- The system shall allow data restoration in case of system failure.
- The system shall use RabbitMQ to design the event bus using the Erlang and Advanced message
- The system shall allow authorized users to access the interface within 5 seconds after logging in.
- The system shall automatically move reports older than 24 months to low-cost archival storage.
- The system shall allow exporting reports to PDF format.
- The system shall allow exporting reports to CSV format.
- The system shall allow exporting reports to XLSX format.
- The system shall generate all monthly reports within 15 minutes of the first day of each month.
- The system shall work on Windows 11, MacOS 15 and later versions
- The system shall run on major web browsers (Chrome, Firefox, Edge, Safari)

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- The system shall be connected to the central database, updating the system regarding the hospital's requirements.
- The system shall be accessible on tablets, desktops and mobile devices.

4. Use Case Diagram



5. TRACEABILITY TABLE

ID	USER REQUIREMENT	FORWARD TRACEABILITY
U1	The user (donor) shall be able to create a profile account.	A1, A2, A3, A4, A5, A6, A7, A8, A12
U2	The user(donor) shall be able to update their user profile.	A9, A10, A11, A15
U3	The user(recipient) shall be able to only view his/her medical records.	E2
U4	The user(donor) shall be able to donate blood.	A17
U5	The user(recipient) shall be able to input their credentials.	B1
U6	The user(recipient) shall be able to request blood.	B3
U7	The user(recipient) shall be informed of request status.	B4, B5
U8	The user(recipient) shall be able to check instantly if compatible (transfusable) blood is available or not.	B5, B6
U9	The user (recipient) shall be able to check the location of nearest available donor if none of the inventories have matching blood.	B8, B9
U10	The user (recipient) shall be routed to the nearest inventory which has matching blood.	B8, B9
U11	The user (Admin) shall be provided with a report for all donors.	D1, D2
U12	The user (Admin) shall be provided with a report of all recipients.	D3, D4
U13	The user (Admin) shall be provided with reports for blood units.	D5, D6, D7
U14	The user (Admin) shall be provided with reports for	D10, D11

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	each blood donation campaign.	
U15	The user (donor, recipient & Admin) shall be allowed to create their own login credentials.	E6
U16	The user (donor, recipient & Admin) shall be protected from data loss.	E8, E7, A16, A17, E1, E2

ID	SYSTEM REQUIREMENT	BACKWARD TRACEABILITY
A1	The system shall prompt the user to enter his name to create an account.	U1
A2	The system shall prompt the user to enter his address.	U1
A3	The system shall prompt the user to enter his gender.	U1
A4	The system shall prompt the user to enter his date of birth.	U1
A5	The system shall prompt the user to enter his (National ID) NID.	U1
A6	The system shall prompt the user for a loginID for his account.	U1
A7	The system shall prompt the user for a password for his account.	U1
A8	The system shall send all the user's input details to the database.	U1
A9	The system shall prompt the registered user for their loginID and password to login to their account.	U1
A10	The system shall send the credentials to the database to request access.	U2
A11	The system shall display error messages if access is not granted.	U2

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A12	The system shall display error messages for empty records (such as not entering blood group, address, eligibility)	U1
A15	The system shall allow the user to modify their user details if request is granted.	U2
A17	The system shall not allow the modification of medical records by the user.	U16
B1	The system shall allow hospitals and clinics to register their credentials.	U5
B3	The system shall let users create new blood requests, taking into consideration blood type, quantity and urgency.	U6
B4	The system shall validate requests before submission	U7
B5	The system shall allow the viewing of requests in real time.	U7, U8
B6	The system shall send a notification to users based on their request state.	U8
B8	The system shall check the inventory for the availability of blood.	U10
B9	The system shall match blood units automatically.	U10
D1	The system shall generate an annual report for all donors recorded.	U11
D2	The system shall generate an annual report for all active donors.	U11
D3	The system shall provide a report of all recipients recorded.	U12
D4	The system shall provide a report for all active recipients.	U12
D5	The system shall provide a report of the total	U13

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	blood units collected monthly.	
D6	The system shall provide a report of the total blood units utilised monthly.	U13
D7	The system shall provide a report of the total blood units discarded monthly.	U13
D11	The system shall generate a report for each blood donation campaign detailing the number of donors.	U14
D12	The system shall generate a report for each blood donation campaign detailing the total blood collected by type.	U14

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